

SUBHRA PRATIM MONDAL

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ABOUT ME

Motivated developer proficient in React, Node.js, PostgreSQL, and MongoDB, with proven ability to deliver scalable full-stack solutions integrating Machine Learning and Python backends. UGC NET June 2026 qualified - eligible for Assistant Professorship.

SKILLS

- **Programming Languages:** C, Java and Python.
- **Front-End Web Development:** HTML5, CSS3, JavaScript, React.js, Tailwind CSS, Bootstrap.
- **Back-End Development:** Node.js, Express.js, Flask, FastAPI.
- **AI & Machine Learning:** LLMs, RAG, Text Embedding's, Prompt Engineering.
- **Databases:** MongoDB, MySQL, PostgreSQL, ChromaDB.
- **Tools:** GitHub, VS Code, Postman, Dev C++, Eclipse, XAMPP.
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems.

EDUCATION

UNIVERSITY OF CALCUTTA	Kolkata, West Bengal
• Master of Technology in Computer Science and Engineering	CGPA: Pursuing 2026-2028
UNIVERSITY OF CALCUTTA	Kolkata, West Bengal
• Master Of Science in Computer Science	CGPA: 7.80 2024-2026
ASUTOSH COLLEGE UNIVERSITY OF CALCUTTA	Kolkata, West Bengal
• Bachelor Of Science in Computer Science	CGPA: 7.61 2021-2024

PROJECTS

[AUTOPARKIQ](#) - Automated Car Parking System (React, Node.js, Python, ML)

- Developed a machine learning model analyzing real-time data to forecast parking availability, improving space utilization by 25%.
- Integrated a Python-based backend logic for data processing with a responsive React frontend.

[OPTIRoute](#) - Adaptive Traffic Signal Benchmarking System (SUMO, Python, React, Flask)

- Built an adaptive traffic signal control algorithm and benchmarked it against baseline strategies using SUMO microscopic simulation, validating results with statistical significance testing.
- Engineered a real-time digital twin dashboard integrating React and Flask to visualize simulation outcomes, improving performance analysis clarity by 20%.

[SATHI-SOS](#) - Accident Detection & Emergency Response App (React Native, .NET, PostgreSQL)

- Built a multi-sensor accident detection system fusing accelerometer, GPS, and audio data to auto-trigger SOS alerts with confidence-based logic.
- Implemented real-time hospital dispatch using SignalR and PostGIS geolocation queries, with secure JWT authentication for multi-user driver accounts.

PROFESSIONAL EXPERIENCE

Main Coder Intern Fr Global Macro Private Limited	April 2023 - Aug 2024
• Spearheaded architectural planning and agile initiatives across a 5-member team, streamlining workflows and reducing sprint delays by 20%.	
• Successfully delivered 5 major full-stack web applications on schedule using React, Node.js, MongoDB, and PostgreSQL, boosting overall team productivity by 25%.	

ACHIEVEMENTS

- Qualified UGC NET - Assistant Professor Eligibility. June 2026
- Participate in the Ninth International ICDMAI hackathon. 22 Dec 2024
- Class Representative & Placement Coordinator - University of Calcutta. 2024–2026